DAT540 Project Guidelines

The objective of group projects in this course is to solve problems by working with real-world datasets. The aim is to think like a data scientist and go through all the steps of a typical data science project.

During this project, the student will learn about designing a research plan, preprocessing data, preparing input for the model, selecting an algorithm and training the model, interpreting the results, and evaluating and reporting the outcomes scientifically.

General Rules:

- Every project has a distinct dataset, and you must perform the exploratory analysis necessary for the final task.
- Each group may consist of **2–5 students** (i.e., two, three, four, or five students). In exceptional cases, **individual students may be allowed to work alone**, subject to approval. If you have a specific reason for working individually or need to discuss your group arrangement, please contact Mina.
- Sign up for the group on the spreadsheet. A group leader is assigned within each group and is responsible for teamwork and coordination.
- You can contact the TAs to meet him/her for refined consultation.

Project Difficulty and Grading

- All projects will be considered equivalent in terms of difficulty for grading purposes. Choosing a more advanced project will not give you additional credit or a higher grade.
- However, if you would like to challenge yourself and explore a more advanced topic, you are welcome to choose one of the advanced projects.

General Assessment Criteria:

1. Did the group complete all the steps of the data science project?
2. Did the group formulate a valid research question?
3. Did the group use the data to try to answer the research question? Is the presentation self-contained?

4. Is the notebook submission fully explained and well-documented? Did the entire group contribute to the task?
5. Does the work cite relevant sources, and are the authorities acknowledged correctly?
6. Do the report and presentation clearly explain the various aspects of the project and justify the decisions made?

Key Notes for Oral Examination:

• Maximum Duration: 10 minutes

• Focus on demonstrating and explaining the code, not repeating the written report

• Show how the code works, what each major block does.

The table below outlines the guidelines for evaluation of your data science project in more details. It covers three delivery components and associated tasks: Code/Notebook, Written report, and Oral presentation.

Componenent	Task	Description	Evaluation Points
Notebook/ Code files	Data Exploration	Initial Data Review	Shape of dataset; data types; feature overview; missing values; duplicates; etc.
	Data Exploration	Cleaning	Handled nulls; removed duplicates; addressed inconsistencies; etc.
	Data Exploration	Visualization	At least 3 meaningful plots; insights clearly interpreted
	Data Exploration	Preprocessing	Type conversions; normalization or scaling; outlier handling when needed
	Data Exploration	Feature Selection	Identified key features; justification provided
	Model Selection	Model Justification	Explained suitability; aligned models with problem
	Model Selection	Model Variety	At least 3 models tested
	Model Selection	Research Question	Clearly defined problem or hypothesis
	Training	Data Preparation	Split data correctly; normalization applied if needed; nulls handled
	Training	Training Performance	Reported accuracy/error/metrics interpreted
	Validation Evaluation	Evaluation Metrics	Metrics used for classification/regression/clustering (for example: accuracy, regression error, Silhouette Coefficient)

	Validation Evaluation	Parameter Tuning	Evidence of tuning or optimization if needed
	Validation Evaluation	Predictions	Tested on unseen data; showed sample outputs
Report		Structure & Clarity	Clear sections; concise writing; max 4 pages
Report		Problem Definition	Dataset described; objective clearly stated
Report		Method Summary	Key preprocessing and models summarized
Report		Results & Insights	Performance summarized; insights explained
Report		Conclusion	Findings; challenges; future improvements
Presentation		Code Walkthrough	Explain core parts of the code; demonstrate how each part works |
Presentation		Implementation Decisions	Justify coding choices (functions, libraries, preprocessing steps) and why they were implemented that way
Presentation		Model Coding & Training	Show how models were implemented in code, including training loops and parameter settings
Presentation		Interpreting Outputs	Explain evaluation results directly from code execution (plots, predictions)